

JOSHUA COWELL

Active Top Secret / SCI Clearance | Full-Scope Polygraph

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EDUCATION

Bachelor of Science in Mechanical Engineering

Expected May 2027

Liberty University, Lynchburg, VA

GPA: 3.85

Selected Coursework: Dynamics, Fluid Dynamics, Heat Transfer, Materials Engineering, Differential Equations, Thermal-Fluids Laboratory

Founding Member: American Nuclear Society (ANS) student chapter (2025); AIAA student chapter (2026)

Leadership: Liberty Rocketry – Chairman, Board of Advisors (Jul 2026 – Present)

ENGINEERING EXPERIENCE

Advanced Weapons Systems Analyst Intern

May 2026 – Present

U.S. Government

- Conduct physical testing of engineering systems and analyze results to calibrate simulation models
- Developed a Python-based geospatial analysis tool to meet team requirements, using AI-assisted development tooling to iterate on the design
- Perform technical assessments of advanced engineering systems alongside multidisciplinary teams, translating findings into decision-ready summaries
- Brief senior leadership and author concise written products that communicate complex technical data clearly
- *Additional detail available upon request*

Chief Systems Engineer and Team Lead

Jun 2025 – Jul 2026

Liberty Rocketry

Earlier roles: Assistant Chief Engineer; Senior Structural Engineer; Fin Fluid-Dynamics Engineer

- Placed 18th of 141 international collegiate teams at the 2026 IREC
- Led a 60-member student team in the design, testing, and launch readiness of a high-powered rocket
- Oversaw engineering decisions across propulsion, avionics, recovery, and aerodynamics subsystems
- Facilitated communication among internal teams, external vendors, and university faculty to improve project execution
- Reviewed SolidWorks, ANSYS, MATLAB, OpenRocket, and RocketPy models and simulations used for team design decisions

Thermal Hydraulics Engineering Intern

May 2025 – Dec 2025

Framatome Nuclear

- Developed and validated a Python predictive-wear model against documented component wear, reducing prediction error by 75% versus the legacy model
- Built a Python/xarray algorithm processing six-dimensional datasets to calculate safe operating limits, reducing analysis time from 800+ hours to minutes
- Performed system-level thermal-hydraulic and FIV analyses using AFT Fathom and custom tools to support component qualification
- Authored technical documentation and presented findings to engineering leadership; received a \$2,500 technical-presentation award

Lead Undergraduate Research Assistant

Sep 2024 – Present

Liberty University

- Lead a five-person team conducting split-Hopkinson pressure bar (SHPB) testing for dynamic material characterization
- Develop test procedures and fixtures; use PicoScope strain data to generate stress-strain curves and correlate high-speed imaging with SEM results

TECHNICAL SKILLS

Systems & Test: System integration, test planning, model calibration, subsystem coordination, technical documentation

Analysis & CAE: ANSYS Fluent, ANSYS Mechanical, AFT Fathom, OpenRocket, RocketPy, CFD, FEA

CAD & Manufacturing: SolidWorks, Onshape, GD&T, technical drawings, 3D printing, CNC machining

Programming & Lab: Python (NumPy, pandas, xarray), MATLAB, SHPB testing, SEM, high-speed imaging

SELECTED PROJECTS

Unmanned Aerial Vehicle Design & Analysis

Tools: SolidWorks, ANSYS Fluent, 3D Printing

- Designed and integrated a modular UAV airframe, propulsion system, and payload using replaceable 3D-printed components; evaluated performance with CFD

3D-Printed Model Rocket

Tools: ANSYS Fluent, OpenRocket, Additive Manufacturing

- Engineered a reusable threaded airframe with a hybrid recovery system; simulations predicted Mach 0.5+ and approximately 3,000 ft altitude